



OPEN Decoding covert visual attention of electroencephalography signals using continuous wavelet transform and deep learning approach

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Covert visual attention decoding from EEG signals is a key challenge in cognitive neuroscience and brain–computer interface applications. Traditional approaches often rely on manual feature extraction and handcrafted pipelines, which limit scalability and generalization. In this study, we propose a deep learning-based framework that leverages time-frequency representations, specifically Continuous Wavelet Transform (CWT), to enable end-to-end classification of covert attention states without manual feature engineering. EEG data were recorded from ten healthy participants performing spatial and feature-based attention tasks. Among the tested models, ShallowConvNet achieved 100% accuracy in binary classification and over 90% in four-class conditions. EEGNet also performed competitively, exceeding 97% and 88% accuracy in two- and four-class scenarios, respectively. These findings demonstrate that integrating CWT with deep neural architectures significantly enhances decoding performance compared to conventional raw-signal approaches, offering a scalable and efficient solution for real-time attention monitoring.

Keywords Electroencephalography, Visual attention, Deep learning, Shallow convolutional neural networks, Electroencephalography convolutional neural network, Continuous wavelet transform

Brain-computer interfaces (BCIs) facilitate communication and the operation of assistive devices for individuals with significant paralysis. Since most BCIs rely on gaze shifts, they are often ineffective for users with limited eye movement. To address this challenge, researchers have worked on designing BCIs that operate without depending on eye movements¹.

The ability to choose and prioritize visual information according to our objectives is made possible by attention, which allows us to process more information than our eyes can handle in the visual world. Our attention typically follows our eye movements and is concentrated on our central vision². It is also possible to covertly align it in space without shifting our gaze; this is known as “looking out of the corner of our eye.”³. It is believed that covert visual attention serves as a quick way to search the field of vision for needed information. The use of covert visual attention in BCI applications has been proposed in numerous studies^{4,5}. Focusing on specific areas or characteristics allows visual attention to emphasize and selectively process certain pieces of information while disregarding others⁶. Covert visual attention (CVA) plays a vital role in daily life by allowing us to: Monitor surroundings without obvious gaze shifts (e.g., checking traffic while focusing ahead), Process peripheral information during social interactions (e.g., observing others while maintaining eye contact), Rapidly scan visual scenes when overt eye movements would be inefficient (e.g., locating objects in cluttered environments).

The visual attention mechanism enables the nervous system to prioritize specific locations, features, or entire objects, giving rise to spatial, feature-based, and object-based forms of attention. Spatial attention refers to the allocation of processing resources to particular locations in the visual field, thereby enhancing the perception of stimuli at those locations. Baldauf and Deubel introduced the concept of “attentional landscapes,” demonstrating how attention is dynamically distributed in space to support visuo-motor coordination⁷. Similarly, Bagherzadeh

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et al. provided causal evidence that oscillatory alpha activity can be modulated via neurofeedback to influence spatial attention allocation⁸.

Beyond spatial dimensions, feature-based attention enhances sensitivity to specific stimulus attributes—such as colour, orientation, or motion—regardless of their spatial location. Feature-based attention allows the visual system to prioritize specific stimulus attributes—such as motion direction or colour—regardless of their spatial location. This mechanism enhances perceptual sensitivity by modulating neural responses in a feature-selective manner. Treue and Martinez-Trujillo found that attending to a particular direction of motion led to increased neural activity in motion-sensitive areas of the visual cortex, independent of where the stimulus appeared in space⁹. Consistent with this view, Schwedhelm et al. demonstrated that the lateral prefrontal cortex of non-human primates encodes both the colour of a stimulus and its task relevance, suggesting a feature-based attentional modulation in higher-order cortical areas¹⁰.

Object-based attention, in contrast, refers to the prioritization of whole perceptual objects rather than isolated features or locations. Baldauf and Desimone demonstrated using fMRI that attention can preferentially enhance responses to behaviorally relevant objects, even when they are spatially separated, suggesting that attentional selection can operate at the level of integrated object representations¹¹.

Electroencephalography (EEG) is a neuroimaging technique that is non-invasive, portable, and economical, making it popular for the development of BCIs¹. Many effective BCI systems rely on visual attention and use EEG signals, such as oscillations and evoked responses, to assess the user's attentional state when instructed to focus on a specific visual stimulus. These visual models can achieve high transmission efficiency and decoding accuracy because of the robustness and strength of visual responses in EEG¹². Additionally, EEG is frequently utilized to examine the timing of brain activity involved in completing cueing tasks. Specifically, event-related potentials (ERPs) can be used to investigate brain responses associated with shifts in visuospatial attention¹³.

Learning discriminative signal patterns from brain data can be accomplished effectively with machine learning techniques. Feature extraction is a prerequisite for classification in the majority of traditional machine-learning methods. The process of feature extraction is guided by specific hypotheses and necessitates expertise in the relevant domain. Using a machine learning algorithm to automatically identify relevant features can save time and effort. This algorithm may also uncover and utilize features that would have been missed or eliminated. In recent years, deep learning, which is a subset of machine learning, has surfaced as a technique for automatically identifying features in data that enhance the ability to differentiate between classes. Deep learning has demonstrated efficacy across a range of tasks, particularly in image and audio data¹⁴.

Trachel et al. utilized the common spatial patterns (CSPs) algorithm to extract power features in the alpha-band that effectively differentiate the attended location in individual trials. A linear support vector machine (SVM) with regularization was employed to interpret the individual trial epochs¹⁵. Thiery et al. developed a machine-learning classification pipeline by extracting the extensively researched ERP components (N2pc - SPCN), and then using those features as inputs¹³. Ahmadi et al. employed phase transfer entropy (PTE), and the findings indicate that the alpha and beta1 frequency bands yield superior performance in comparison to other frequency bands after selecting the optimal decoding feature¹⁶. Desantis et al. investigated whether raw channel activity and alpha power recorded during the cue-response interval could both predict the orientation and temporal dynamics of covert attention using multivariate pattern classification analysis². Schneider et al. studied the topic of continuous closed-loop feedback in a covert visuospatial attention (CVSA) task using the occipital α -power lateralization¹⁷. The decoding approach developed by Reichert et al. involved a correlation measure and data-driven estimation of spatial filters to decode the N2pc¹. Davoudi et al. utilized features from cross-frequency coupling (CFC) measures to differentiate EEG signals associated with upward or downward directions, as well as distinguishing between blue and orange colours. They employed several classification methods, including SVM, Extreme Learning Machines (ELM), and Quadratic Discriminant Analysis (QDA)⁶. Kosmyna et al. in their study showed how to set up an AR EEG-BCI integrated prototype online and in real-time using CVSA without the need for external stimulation¹⁸. Rezaei et al. employed a graph theory-based approach on EEG signals, leveraging coherence features. They successfully decoded covert visual attention using QDA, emphasizing the efficacy of graph-theoretical features like node degree and the role of beta and gamma frequency bands¹⁹.

Traditional machine-learning techniques are commonly used in the analysis of EEG signals. However, the accuracy and performance of EEG signal processing are inadequate. Researchers are exploring the potential of different deep-learning methods for analyzing and interpreting EEG signals to tackle this issue.

To decode covert spatial shifts, deep learning may be a useful technique. It assists with building a strong framework that works better than earlier techniques²⁰.

Although CVA has been widely studied using EEG, most prior research has focused on limited task types and conventional analysis methods, often overlooking how attention type and signal representation affect decoding performance. In particular, systematic comparisons of spatial and feature-based attention using deep learning, and assessments of how input representations like raw EEG or CWT influence classification accuracy, remain scarce. Addressing this gap, the present study evaluates multiple deep neural architectures across different signal formats and CVA paradigms, aiming to uncover how these factors interact in shaping decoding performance.

The rest of this article is organized as follows: Sect. 2 describes the dataset, experimental paradigms, EEG recording, preprocessing, and the deep learning models along with time-frequency transformation methods. Section 3 presents the classification results for both binary- and multi-class cases, comparing the performance of different models and input representations. Finally, Sect. 4 discusses the findings, their implications, and concludes the study.

Methods

Examining participants engaged in a covert visual attention task provides valuable insights into neural activity patterns associated with attention across the brain. To explore the impact of interactions among brain oscillations on the processing of differences in colour, luminance, and motion direction, we used 20-channel EEG data gathered from healthy participants. Decoding accuracy was evaluated through time-frequency (TF) analysis, convolutional deep-learning models, and a 5-fold cross-validation approach. Detailed descriptions of these methods are presented in the following sections. The approach adopted in this research is shown in Fig. 1.

EEG recording

EEG data were recorded from 15 participants (mean age: 26.6 ± 1.3 standard deviation) while they performed covert visual attention tasks. All participants were comfortably seated in soundproof environments, positioned in armchairs, with their gaze directed forward, maintaining a distance of 70 cm from 21-inch computer monitors. They were instructed to remain still and have either normal or corrected-to-normal vision during the recordings. The EEG dataset⁶ used in this study was previously collected in accordance with the Declaration of Helsinki and approved by the Neuroscience & Neuroengineering Research lab. ethics committee of IUST. All participants provided written informed consent before the study. The data were fully anonymized before access in the first months of 2023, and no additional data collection was performed. The process of signal recording for covert attention to colour is illustrated in Fig. 2.

EEG signals were recorded using the international 10–20 system, employing a total of 20 channels: FP1, FP2, F3, F4, Fz, F7, F8, T3, T4, C3, C4, Cz, T7, T8, P3, P4, Pz, O1, O2, and Oz. These channels were referenced to the average voltage of electrodes placed on both earlobes. The data were sampled at a frequency of 250 Hz, amplified with specialized hardware, and then digitized using an analog-to-digital converter.

More information about the data can be found in⁶.

EEG data analysis

All data were examined using EEGLab²¹ and MATLAB (R2017a). After the raw data was filtered with a FIR filter spanning 1–60 Hz, a notch filter was then applied to remove 50 Hz line noise. The lower cutoff of the FIR filter was set at 1 Hz to eliminate the DC offset in the signals. Independent components analysis (ICA) was used to remove artifacts, including EOG, eye movements, ECG, and unwanted cranial muscle activity. After artifact removal, the continuous signal was divided into trials, each starting with the presentation of the stimuli and ending with their first change. Only accurate trials were chosen for additional examination.

Behavioral task

Participants completed covert visual attention tasks that targeted either colour or direction, which were created using Matlab with Psychtoolbox Version 3 (PTB-3). There were 96 trials in total, with participants allowed 10-minute breaks between sessions. To maintain covert attention and prevent direct eye contact with the stimuli, participants were instructed to focus their gaze on a specified point on the screen. A cue was presented 450 milliseconds after the trial began and remained on the screen for 650 milliseconds.

In the colour task, stimuli included orange and blue dots positioned on the left or right side of the fixation point. A horizontal cursor, which could be blue or orange, served as a cue to indicate the target side and colour for covert attention. Participants needed to focus on both the side (left or right) and the colour (orange or blue) of the cursor. They responded by pressing the space key whenever a change in the luminance of the target stimuli was detected.

In the direction task, participants were shown two groups of moving dots—one group moving upward and the other downward—positioned on both sides of the fixation point. A diagonal white cursor served as the cue, indicating the direction (up or down) and the side (left or right) of the fixation point to be attended to. Participants had to pay attention to both the direction and side specified by the cue and press the space key whenever they observed a change in the target stimuli's direction.

Changes in both colour luminance and motion direction were introduced unpredictably, with the timing of target variations differing across trials to sustain participant attention throughout the task. Only trials in which participants provided accurate responses were considered for further analysis. A visual representation of the experimental paradigm can be seen in Fig. 3.

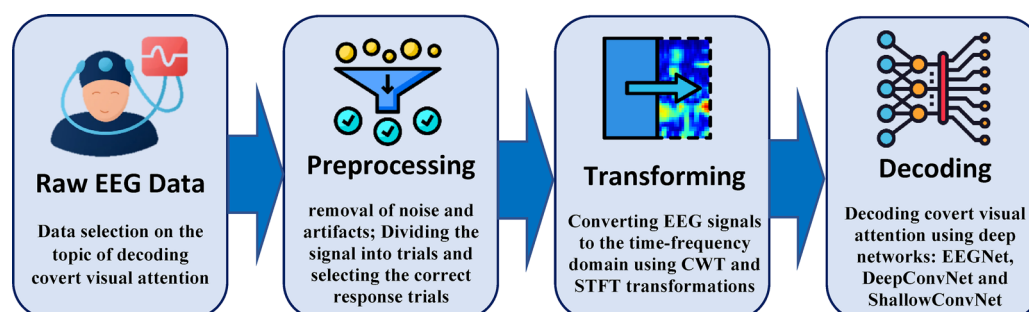


Fig. 1. The schematic representation of the procedures that have been executed.

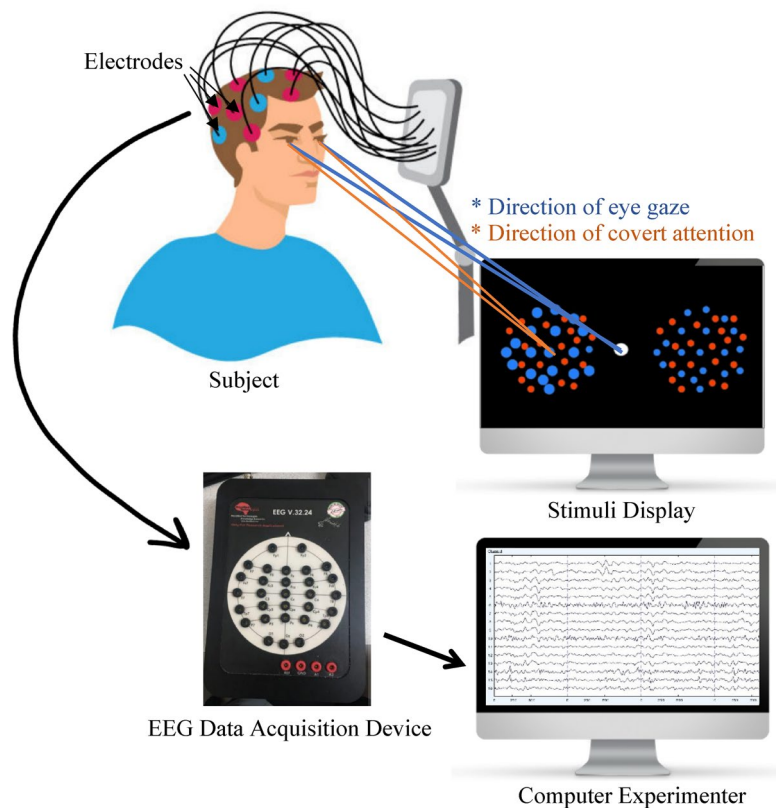


Fig. 2. Procedure for recording during the covert attention to colour task. EEG was utilized to record brain activity during the completion of two types of tasks, focusing on colour or direction.

Time-frequency analysis

Although the Fourier Transform (FT) can analyze a wide range of signals, many real-world processes—including brain activity—are non-stationary. Signals representing these processes can have time-varying frequency and amplitude, but such changes cannot be captured by the FT, as it integrates the signal over the entire time domain. By assuming a signal is stationary over time, the FT eliminates the rich temporal information present in EEG data. Additionally, when a signal is non-stationary, the precision of the FT is compromised²².

Short time Fourier transform

Non-stationary signal analysis requires additional techniques. The Short-Time Fourier Transform (STFT) is one such technique. By using this method, the signal is convolved with a window function that has a center and non-zero value within a limited interval around it. The outcome of this convolution is then subjected to the FT.

The resulting Power Spectral Density (PSD) corresponds to the frequency spectrum centered around the time when the window is positioned. By moving the window along the time axis, we observe how the power at different frequencies evolves over time. This process yields a two-dimensional representation of the signal, where time and frequency are divided into discrete 'bins', each indicating the power density at a specific time and frequency²³. By analyzing the STFT spectrogram, we can gain insight into how the different frequency components of the EEG signal change over time and reveal patterns and dynamics related to covert visual attention. It is important to note that the choice of STFT parameters, such as window length, window function, and overlap, can affect time-frequency resolution and spectral leakage. Careful investigation and optimization of these parameters are necessary to ensure accurate and meaningful analysis of EEG signals. The mathematical equation for this conversion is as follows²⁴:

$$\text{STFT}(\tau, \omega) = \int_{-\infty}^{\infty} x(t) w(t - \tau) e^{-j\omega t} dt \quad (1)$$

Where $x(t)$ represents the original EEG signal, $w(\tau)$ denotes the window function (Hanning), τ refers to the time axis, and ω represents the frequency axis. The spectrogram equation of this transformation is given below²⁴:

$$\text{spectrogram} \{x(t)\}(\tau, \omega) = |\text{STFT}(\tau, \omega)|^2 \quad (2)$$

In this study, the *scipy.signal* library and the *stft* module were used to obtain the spectrogram of each channel using the Python programming language in Google Colab.

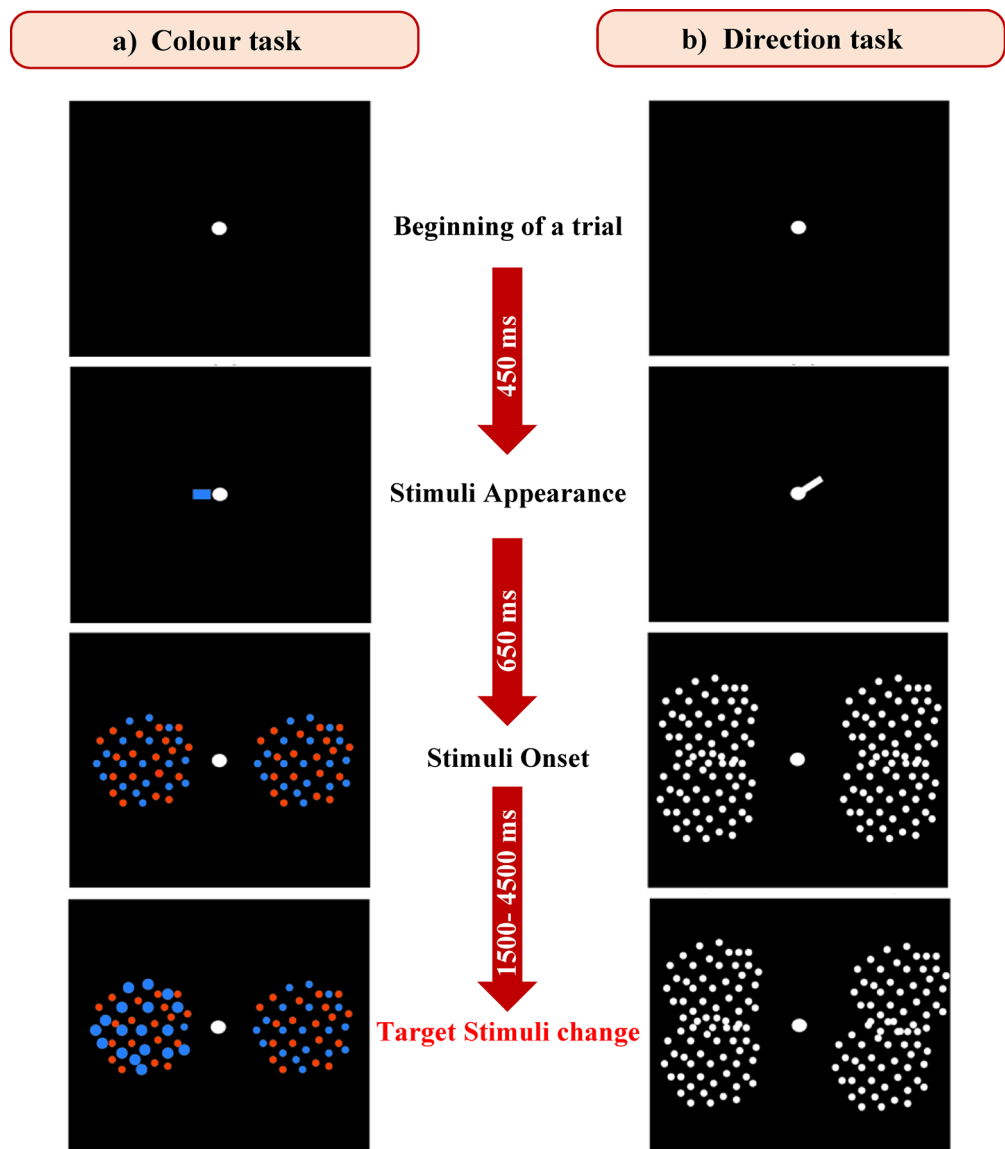


Fig. 3. Experimental procedure patterns for covert visual attention. While the fixation time and cue presentation were fixed, the interval between the stimulus change and the stimulus onset was randomized in each trial. (a) Attention to colour, (b) Attention to direction.

The input matrix, which was three-dimensional with dimensions for trials, channels, and time samples (direction task: $588 \times 20 \times 538$; colour task: $514 \times 20 \times 532$), was transformed into a four-dimensional matrix with dimensions for experiment channels, frequency (frequency intervals), and time (time windows), resulting in dimensions of $588 \times 20 \times 129 \times 6$ for the direction task and $514 \times 20 \times 129 \times 6$ for the colour task.

Continuous wavelet transform

STFT is a tool used to analyze signals over time and frequency. However, it may not be effective for EEG signals because it can't provide high resolution in both time and frequency simultaneously. If the STFT window is small, it gives us a clear picture of when things happen in time but not much detail about the frequencies involved. A large window does the opposite, giving us more information about the frequencies but less about the timing of events. To address this issue, we use the Continuous Wavelet Transform (CWT)²⁵. Like the FT, CWT compares the original signal to a basic pattern, called a mother wavelet, instead of a sine wave. The advantage of CWT is that it can be adjusted to focus on different frequencies at different times, giving us a detailed map of the signal's frequency and timing without the limitations of the STFT²⁶. The mathematical formula of CWT is given in Eq. (3)²⁷:

$$\text{CWT}(\omega, s) = \frac{1}{\sqrt{|s|}} \int x(t) \psi\left(\frac{t-\omega}{s}\right) dt \quad (3)$$

where $x(t)$ is the EEG signal in this paper, ψ is the mother wavelet, ω is a translation or time-shifting parameter, and s is a scaling parameter. Correlation coefficients of CWT are represented by $\text{CWT}(\omega, s)$.

CWT is a method that breaks down a continuous-time signal into many different frequency parts. It does this by stretching and shifting a wavelet, which is like a small wave, and using it to measure the signal. The result of this process is called a scalogram, which is a way to see how the energy of the signal is spread out over time and at different frequencies. Think of it as a detailed map that shows where and when different frequencies of the signal are strong or weak^{28–30}.

The scalogram, similar to the spectrogram mentioned in the previous section, can be utilized in³¹ deep learning applications owing to its visual characteristics. To derive the scalogram for each channel, the *pywt* library and *cwt* module were employed using the Python programming language in Google Colab, with the Morlet wavelet function as the mother wavelet. The equation for Morlet's mother wavelet function is as follows²⁷:

$$\psi_{\text{Morl}}(t) = \frac{1}{\sqrt{\pi}f} e^{j2\pi f t} e^{-t^2/f} \quad (4)$$

A logarithmic scale ranging from 1 to 50 Hz, consisting of 25 components, was employed. This facilitated the transformation of the data into a four-dimensional structure with dimensions corresponding to trials, channels, time, and frequency (scale). Following the transformation, the dimensions for the direction task were $588 \times 20 \times 538 \times 25$, and for the colour task, they were $514 \times 20 \times 532 \times 25$.

Using deep learning models as a decoder

In the current research, we utilized three types of deep learning models commonly employed for EEG signal classification: EEGNet³², ShallowConvNet, and DeepConvNet³³. Unlike most research that relies solely on the raw EEG signal, the input for these models was the transformed time-frequency domain data obtained through STFT and CWT transformations. All models were trained on Google Colab using a GPU within the TensorFlow framework³⁴, using the Keras library³⁵. The batch size and the number of epochs were empirically determined to be 32 and 50, respectively, for all models, and Adam optimizer was selected to fine-tune the model parameters.

EEGNet

EEGNet is a dense convolutional neural network introduced in 2018 by Lawhern et al. for EEG-based brain-computer interfaces. This architecture is capable of: (1) being utilized within various BCI paradigms, (2) operating effectively with minimal training data, and (3) generating features that provide meaningful neurophysiological interpretation³².

First block At the outset of the two convolutional stages, a two-dimensional convolutional filter F1 with a size of (1,32) is employed. Subsequently, a depth convolution with the size of (20,1) is utilized to learn a spatial filter (where 20 represents the number of channels). In computer vision applications using Convolutional Neural Networks (CNNs), the primary advantage of employing deep convolutions lies in reducing the number of trainable parameters that need adjustment. This reduction occurs because these convolutions are not fully connected to all previous feature maps. Importantly, when applied to specific Electroencephalogram (EEG) tasks, this operation directly facilitates the learning of spatial filters for each temporal filter. The design of this two-stage convolutional sequence is partly inspired by the FBCSP (Filter Bank Common Spatial Pattern) algorithm. Before applying the Exponential Linear Unit (ELU) nonlinear activation function, batch normalization is performed along the feature map dimension after each convolution. Additionally, the Dropout technique is employed to regularize the model, with a dropout probability set to 0.5 to mitigate overfitting when training on a small sample size. Preceding these steps, an average pooling layer is applied to reduce the computational volume.

Second block In the initial step, a separable convolution is employed, which consists of a deep convolution followed by a pointwise convolution of size (1,1). The main advantages of separable convolutions are twofold: (1) they reduce the number of parameters that need adjustment, and (2) they explicitly separate the relationships within and between feature maps. This separation is achieved by learning a kernel that summarizes each feature map individually and then optimally integrates them. When used specifically for EEG applications, this learning operation decouples the process of summarizing individual feature maps in the temporal domain (depth convolution) from the process of optimally combining feature maps (pointwise convolution). Following this, similar to the previous block, a batch normalization layer and a nonlinear activation function are applied. An integration average pooling layer is then used to reduce the dimensions, and finally, the smoothing layer results in a one-dimensional feature map.

Third block In this block, which is the classification block, a dense layer is used to aggregate the features and the data is classified with the Softmax activation function.

A visualization of the EEGNet model blocks can be found in Fig. 4.

DeepConvNet

DeepConvNet consists of four convolutional blocks terminated by a dense classification layer. Specifically, the first convolutional block consists of sequential temporal and spatial convolutions that effectively encode spatial information between EEG channels. The remaining three blocks are made by temporal convolution only. This network, which is deep and contains a large number of parameters, as its name suggests, was first introduced in 2017 by Schirrmeyer et al.³³.

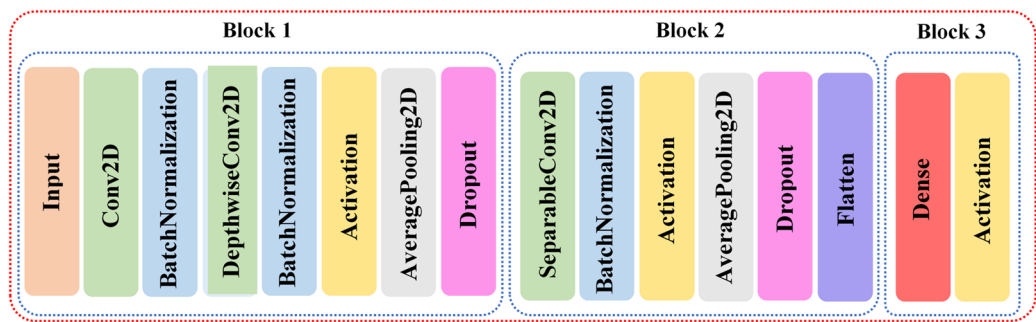


Fig. 4. General structure of the EEGNet model with 3 blocks.

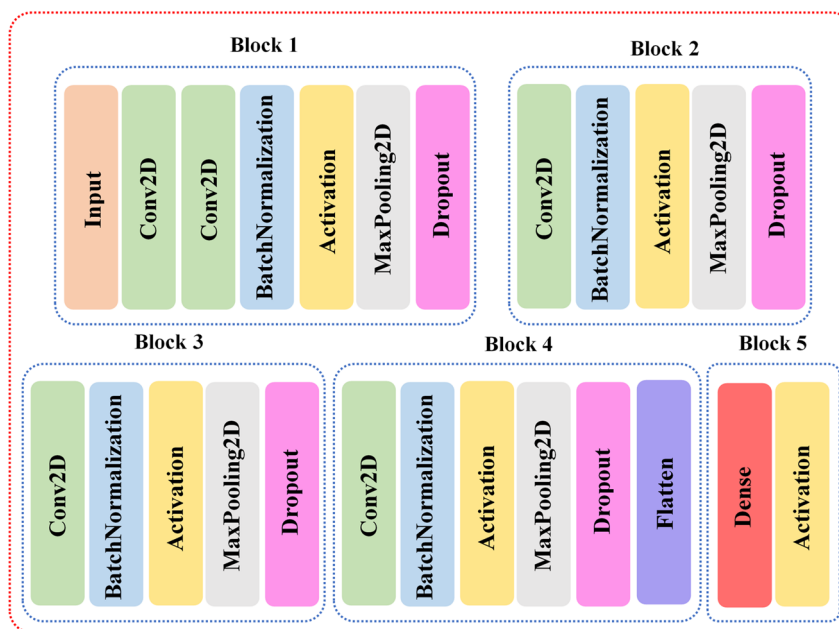


Fig. 5. General structure of the DeepConvNet model with 5 blocks.

First block As depicted in Fig. 5, this model is deeper than the previous model and comprises additional blocks:

In the initial layer, each filter performs a time-based convolution. The second layer applies a spatial filter to all electrode pairs, taking into account the temporal convolutions from the first layer. While there is no activation function between these two layers, they can technically be merged into one. However, the two-layer structure helps to regularize the convolution process by splitting the linear transformation into distinct temporal and spatial components. Then, a batch normalization layer is followed by an activation layer using the Exponential Linear Unit (ELU) function. Subsequently, a maxpooling layer is employed. Finally, this block concludes with a Dropout layer, where the dropout probability is set to 0.5.

Second block, third block, and fourth block All three of these blocks consist of the following components: a convolution layer, a batch normalization layer, an activation layer using the ELU activation function, a max-pooling layer, and a dropout layer. In the last layer of the fourth block, a flatten layer is utilized to transform the feature map into a one-dimensional representation.

Fifth block In this block, which corresponds to the same classification stage as the previous model, a dense layer and a Softmax activation layer are employed for the final classification of features.

ShallowConvNet

ShallowConvNet is a simplified variant of DeepConvNet, drawing inspiration from FBCSP. Its first block resembles the initial convolutional block of DeepConvNet, but it differs in several aspects, including the use of a larger kernel, an alternative activation function, a distinct pooling method, and a classification block positioned at the end. This network was introduced simultaneously with DeepConvNet in 2017 by Schirrmeyer et al.³³. ShallowConvNet has gained more popularity than DeepConvNet due to its smaller model size³⁶.

As depicted in Fig. 6, the ShallowConvNet model comprises two blocks:

First block The initial two layers perform a temporal convolution and a convolution with a spatial filter. These steps closely be similar to the processes of the spatial bandpass filter and CSP in the FBCSP algorithm.

Following a batch normalization layer, a nonlinear squaring activation layer is applied. Subsequently, an average pooling layer is employed, and a logarithmic activation function is used in the next layer.

This block concludes with two additional layers: Dropout and flatten.

Second block In the classification block, similar to the previous model, a dense layer and the Softmax activation function are utilized for classification.

Results

The results obtained from the EEG signals decoding are presented in two separate sections. Since the decoding process involves both spatial attention (Binary-class case) and its integration with feature-based attention (Multi-class case), the outcomes are further discussed in distinct segments.

Binary-class case results

In the two-class work of this research, we discussed the decoding of covert visual attention to the left and right sides during colour and direction tasks. The input data for the deep learning models used in classification consisted of three main types: original EEG signals, CWT transformations, and STFT transformations. The results of this classification for these three types of inputs will be described below:

The utilization of STFT as input data yielded suboptimal results. The decoding accuracy remained at chance level, suggesting that this kind of input lacked sufficient discriminative information for effectively classifying signals into left and right or blue and orange categories.

About giving main EEG signals as input data, as can be seen in Fig. 7., the EEGNet and DeepConvNet models have relatively similar performance in this section but DeepConvNet has appeared slightly better than the other (Colour task accuracies: 73.33% – 73.59%, Direction task accuracies: 67.87% – 68.76%, respectively for EEGNet and DeepConvNet models). Also, the results show that the ShallowConvNet model was not a suitable model for classification using this type of input data because the results are close to the chance level (Colour task accuracy: 55.64%, Direction task accuracy: 56.63%). In general, this decoding of spatial attention has a higher performance in the colour task compared to the direction task.

The results of the CWT method as shown in Fig 0.7. for the colour and direction tasks show better performance of all three models than the original signal and the STFT transformation method in both tasks. The ShallowConvNet deep learning model, unlike the previous method, which obtained the weakest result using the original signal as input, is trained in the best way using this method and has the highest performance in both colour and direction tasks (Colour task accuracy: 100.00%, Direction task accuracy: 100.00%). After that, the EEGNet model achieved a good performance in classifying these types of inputs (Colour task accuracy: 97.18%, Direction task accuracy: 98.43%). But the interesting point is that although the performance of DeepConvNet is generally better than the previous methods, this model was weaker in the colour task than in the direction task (Colour task accuracy: 78.72%, Direction task accuracy: 93.71%). It seems that due to the complexity of this model, this method did not contain enough information that is needed in the training of this model. Therefore, lighter models such as EEGNet and ShallowConvNet have been more successful.

In Fig. 7., the better performance of the models in the CWT method is clearly evident. The weaker performance of ShallowConvNet compared to the other two models in the raw signal method can be due to the design of this network with the aim of extracting features related to the logarithm of the band power, which can be decoded using the CWT method with higher performance, while the other two models are designed for event-related potentials²⁴.

Table 1 displays the approximate number of trainable parameters for each model across different methods. These parameter counts remained consistent for both types of experiments since the input data dimensions were identical. An increased number of parameters results in a heavier computational load and a more time-consuming decoding process. Notably, in the CWT method, the number of parameters increases as the input dimensions grow larger.

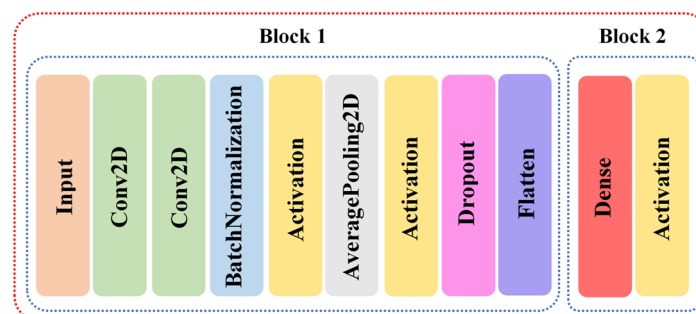


Fig. 6. General structure of the ShallowConvNet model with 2 blocks.

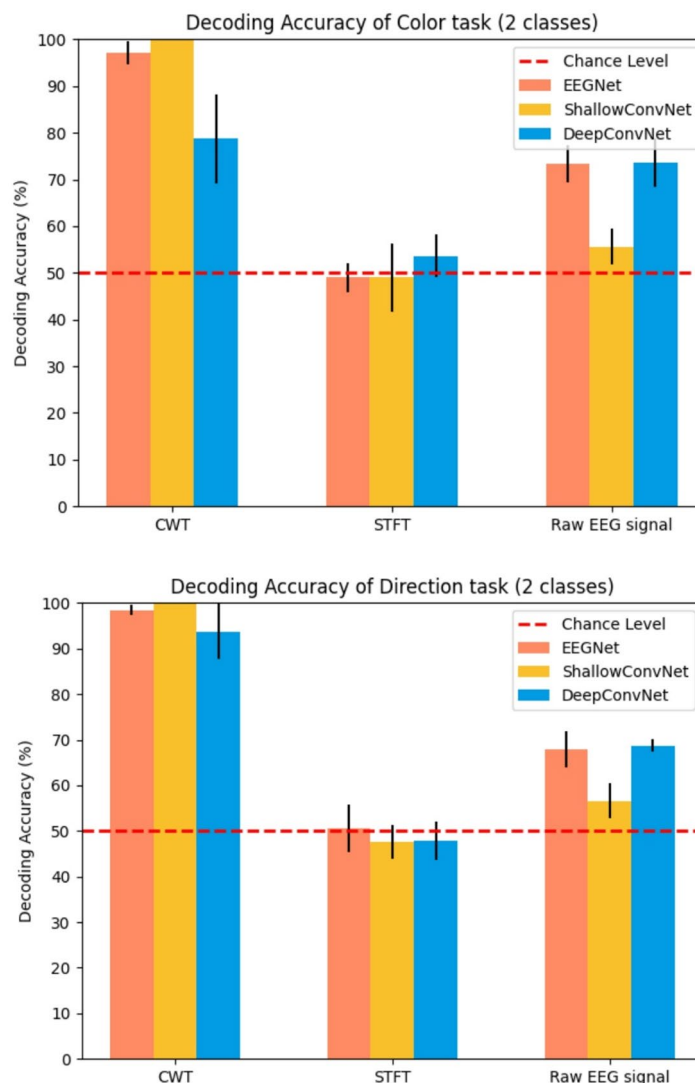


Fig. 7. Comparing the decoding accuracy of three methods in a binary classification study. (a) Colour task (b) Direction task. Salmon colour corresponds to the EEGNet model, Yellow to the ShallowConvNet model, and Blue to the DeepConvNet model. The black lines represent the standard deviation of the results, while the red dashed line indicates the chance level.

Model	Input		
	CWT	STFT	Raw EEG Signal
EEGNet	19,100	6400	5900
ShallowConvNet	115,000	36,200	38,200
DeepConvNet	262,000	147,200	156,600

Table 1. The approximate number of trainable parameters of each model for three methods in the two-class study.

Multi-class case results

In this section, in addition to classifying spatial attention, decoding methods also use colour and direction features for classification. Basically, for the task, the colour classes include: blue-left, orange-left, blue-right, and orange-right, and for the direction task, the classes are up-left, down-left, up-right, and down-right. Then, similar to the binary-class case, in three input methods: main EEG signal, STFT and CWT, their results will be analyzed. Notably, given that this algorithm is four-class, the chance level is 25%. Any performance above this level indicates greater success for that model in classification and decoding.

Similar to the previous section, STFT performed poorly, so we will describe the performance of other methods.

In the multi-class case of main EEG signal method as input, similar to the binary-class case, the DeepConvNet model outperformed the other two models. Specifically, its performance was higher in the colour task (Accuracy: 36.41%) than in the direction task (Accuracy: 33.03%). Also, the EEGNet model performed better in colour task (Accuracy: 34.87%). Unlike them, the ShallowConvNet model exhibited poor performance in classifying this type of input. In fact, in the colour task, its performance was even lower than the chance level. Overall, despite the best available performance, decoding this type of input remained challenging.

In this section of the study, we investigate whether, similar to the binary-class case, the CWT method exhibited the highest decoding accuracy compared to other methods. Specifically, we examine whether it also performs best in the multi-class case. The decoding accuracy displayed in Fig. 8. for colour and direction tasks indicates that all three models outperformed other methods in multi-class decoding (Colour task accuracies: 88.21% – 90.77% – 86.15%, Direction task accuracies: 89.66% – 91.24% – 88.31%, respectively for EEGNet, ShallowConvNet and DeepConvNet models.). Furthermore, their stability, as evidenced by the standard deviation relative to other methods across multiple models runs, is consistently higher. Notably, like the binary-class case, the ShallowConvNet model demonstrated the best performance, while the DeepConvNet model, particularly in the colour task, exhibited the weakest performance. Nevertheless, even considering the 25% chance level, it still achieved desirable performance compared to previous studies in the four-class case.

Figure 8 illustrates side-by-side confusion matrices of the best-performing model (ShallowConvNet - CWT) in two distinct multi-class decoding tasks: colour-orientation (left) and direction-orientation (right). Both matrices exhibit strong diagonal dominance, indicating high decoding accuracy across all classes. In the colour-orientation matrix, the classifier distinguishes between left/right spatial locations combined with colour (blue/orange), with particularly strong performance in “Left Orange” (93.33%) and “Right Orange” (94.67%). Misclassifications are minimal, with slight confusion between “Right Blue” and “Right Orange.” In the direction-orientation matrix, attention states based on directional cues (up/down) show similarly high decoding rates—especially “Right Down” (98.18%) and “Left Down” (95.00%). These results confirm that the decoding framework effectively captures covert attentional shifts, whether driven by spatial-feature conjunctions or directional cues, with CWT-based preprocessing enhancing model performance in both cases.

Finally, in order to check the results of these three methods, their accuracy and the deep learning models used for decoding are compared in Fig. 9 for colour and direction tasks. In this figure, as in the two-class case, the better performance of the models in the CWT method is clearly evident. However, the ShallowConvNet model has been more successful in this regard.

Table 2 shows the approximate number of trainable parameters of each model for each of the methods. These numbers were the same for both types of colour and direction tasks, as in the two-class case, because the dimensions of the input data were not different. More number of parameters means more computational load and a time-consuming decoding process. As it is known, in the CWT method, the number of parameters has increased as the input dimensions become larger. On the other hand, by comparing this table with Table 1, which was related to the two-class case, the result is that the increase in the number of classes is directly proportional to the increase in the model parameters, and the most complex model, the DeepConvNet model, used in the four-class decoding mode, with Applying the transformed input of the EEG signal with the CWT transformation requires training nearly 377,100 parameters in the model.

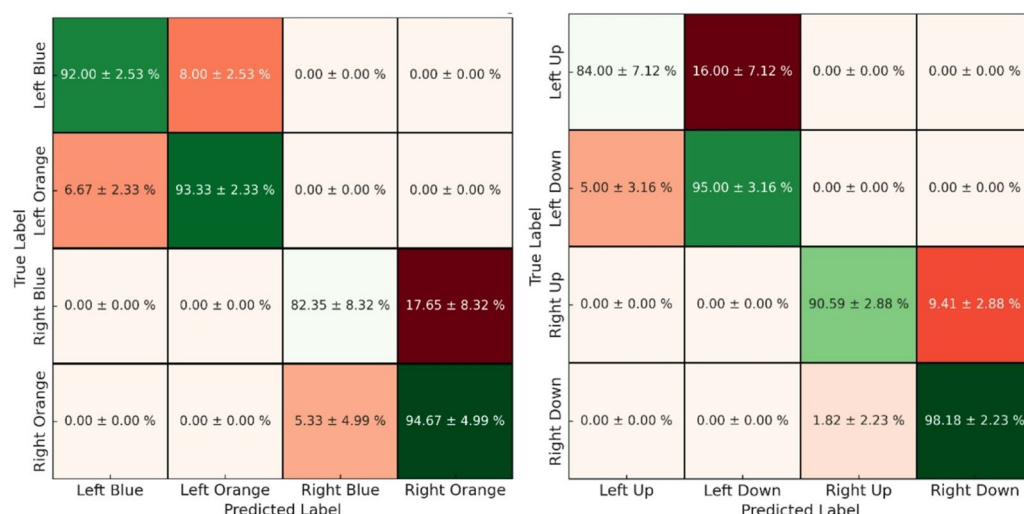


Fig. 8. Side-by-side confusion matrices for the best-performing model (ShallowConvNet - CWT) across two four-class decoding tasks: colour-orientation (left) and direction-orientation (right). Each cell displays the mean classification accuracy ± standard deviation. The prominent high values along the diagonal and minimal off-diagonal confusion indicate the model’s robust capability to accurately decode covert visual attention in both feature-based and spatial-directional contexts.

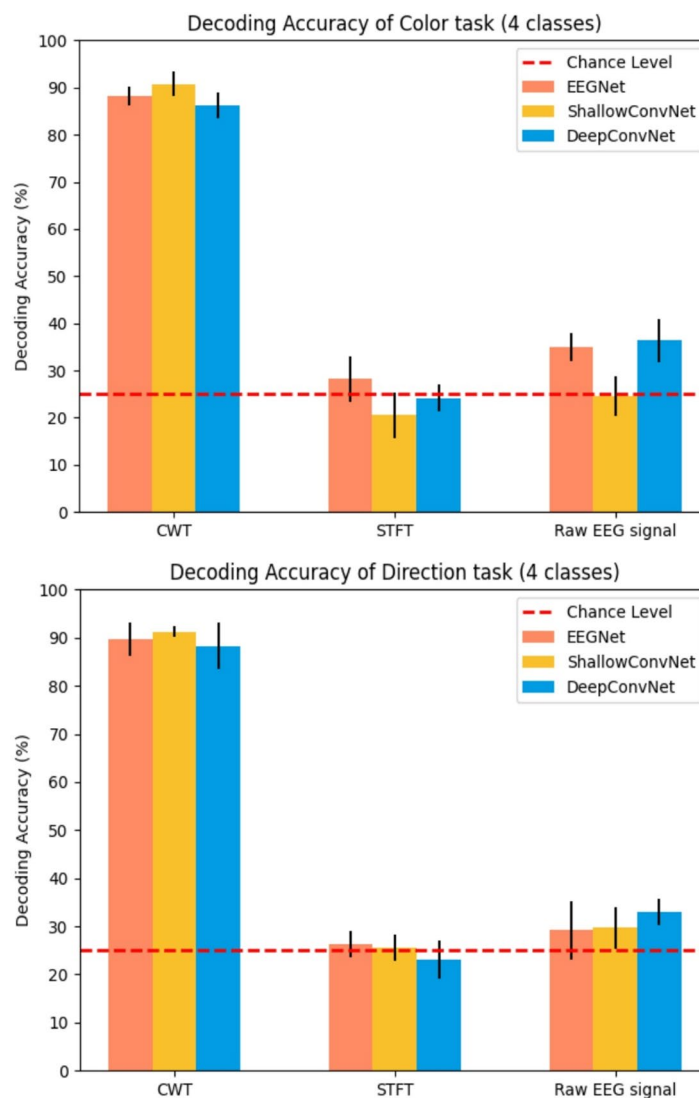


Fig. 9. Comparing the decoding accuracy of three methods in a Multi classification study. (a) Colour task (b) Direction task. Salmon colour corresponds to the EEGNet model, Yellow to the ShallowConvNet model, and Blue to the DeepConvNet model. The black lines represent the standard deviation of the results, while the red dashed line indicates the chance level.

Model	Input		
	CWT	STFT	Raw EEG Signal
EEGNet	31,900	7100	6400
ShallowConvNet	187,000	37,100	44,000
DeepConvNet	377,100	148,800	168,200

Table 2. The approximate number of trainable parameters of each model for three methods in the four-class study.

As shown in Table 3, the proposed approach demonstrates a significant improvement in the ability to decode covert visual attention in both binary and multi-class cases.

Discussion and conclusion

In this research, the decoding of covert visual attention was discussed using brain signals with a deep learning approach. For this purpose, three deep learning models including convolutional layers, which were previously introduced by researchers to decode EEG signals, were used. These models were EEGNet, ShallowConvNet and DeepConvNet, which differed from each other in terms of network depth, number of parameters, type of

Study	Feature extraction method	Classification method	Number of classes	Accuracy (%)	
				Colour	Direction
6	Cross-frequency coupling (CFC)	QDA classifier	2	94.21	90.54
16	Phase transfer entropy (PTE)	QDA classifier	2	91.87	---
			4	65.11	---
19	Graph theory	QDA classifier	2	---	89.96
This study	-----	ShallowConvNet	2	100.00	100.00
			4	90.77	91.24

Table 3. Comparison between the best results of the proposed method and other decoding methods applied to the same data.

feature extraction and activation, and this caused their different performances in this research. The types of input applied to these models were also different so that the network could complete the decoding process with the best performance.

The findings of this study indicate that by converting EEG signals to the time-frequency domain if the chosen transformation has a sufficient resolution in both the time and frequency dimensions, it significantly enhances the ability to decode covert visual attention in existing deep learning models. In this study, both CWT and STFT were utilized. The first transformation (CWT) had the desired capabilities, but the second one (STFT) couldn't fully decode due to the trade-off between time and frequency resolution. It should be noted that using the EEG signals themselves as input in these networks also had a power higher than the chance level for decoding. The best results of decoding for main EEG signals as input belonged to the DeepConvNet model with 73.59 and 68.76% accuracy for the two-class case and 36.41 and 33.03% accuracy for the four-class case for colour and direction tasks, respectively. After that, the EEGNet model achieved good results in this type of input, which the accuracy was 73.33% for the two-class task of colour, 67.87% for the two-class task of direction, and 34.87% for the four-class colour task. This model was not able to decode with high accuracy in the four-class case of direction task with this type of input. The ShallowConvNet model did not get good results with this type of input, while this model achieved the best possible results compared to other models and methods by applying the CWT-transformed input. The results of decoding using this model achieved 100% accuracy in the two-class case for both tasks and also 90.77 and 91.24% accuracy in the four-class case in the colour and direction tasks. In this regard, other models did not decode as much as this model but at a good level. After ShallowConvNet, EEGNet achieved the highest decoding accuracy with this type of input, with accuracies of 97.18 and 98.43% in the two-class case and 88.21 and 89.66% in the four-class case for colour and direction experiments. The DeepConvNet model recorded the lowest accuracy for the two-class case in the colour task with 78.72% accuracy but performed better in the direction task with 93.71% accuracy. The four-class case using this model was associated with 86.15 and 88.31% accuracy, which were acceptable. It seems that this model needs more input data for training to get better results due to its complexity.

It is important to emphasize that the main contribution of this study lies in demonstrating the effectiveness of CWT-based input representations rather than in proposing new network architectures. For this reason, we intentionally adopted three widely used benchmark CNN models (EEGNet, ShallowConvNet, DeepConvNet), so that any observed improvements could be attributed directly to the representation itself. Notably, ShallowConvNet achieved the most outstanding results with 100% accuracy in binary tasks and over 90% in four-class tasks, while being computationally more efficient than DeepConvNet due to its smaller number of parameters and faster training. Nevertheless, we acknowledge that more recent architectures offer promising avenues for future work. In particular, EEG-TCNet³⁷, an accurate temporal convolutional network specifically designed for embedded motor-imagery brain-machine interfaces, represents a promising candidate for future experiments to validate the effectiveness of the proposed CWT representation on state-of-the-art models.

While our study primarily focuses on applying deep learning approaches to EEG signals, particularly leveraging time-frequency transformations such as CWT for enhanced feature representation, our findings can be discussed in the broader context of known neuro-anatomical mechanisms underlying attentional processes. Covert visual attention is widely understood to involve intricate top-down control circuits, notably those orchestrated by the prefrontal cortex (PFC), which are crucial for directing and prioritizing sensory information. For example, existing research by Bedini et al. (2021) provides comprehensive insights into the structural and functional roles of prefrontal regions such as the inferior frontal junction and frontal eye field in top-down control of visual attention, highlighting their involvement in coordinating neural activity underlying cognitive control³⁸. Furthermore, studies on PFC connectivity, such as that by Soyuhos and Baldauf (2023), elucidate the distinct functional connectivity patterns of frontal regions, contributing to our understanding of the neural underpinnings of spatial and nonspatial attention³⁹. The high decoding accuracy achieved by our deep learning models from the CWT-transformed EEG data reflects the model's ability to extract robust electrophysiological signatures generated by these underlying neuro-anatomical networks during covert attentional shifts. Although our methodology does not involve direct anatomical analysis or a visual interpretation of the time-frequency maps, the successful decoding demonstrates that relevant neural information, potentially originating from these known attentional networks, is effectively captured within the transformed EEG data and utilized by our deep learning models. Future work could investigate how the patterns learned by deep networks from time-frequency representations correspond to known neuro-anatomical circuits involved in attentional processing.

Additionally, evaluating the proposed model on multiple datasets would further clarify its generalizability and real-world applicability.

Data availability

The data can be accessed through reasonable request from corresponding author.

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Author contributions

HH: Methodology, Software, Writing – original draft, Writing – review & editing. MRD: Conceptualization, Methodology, Supervision, Writing – review & editing.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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